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Master in Computer Applications

Second Semester

Assignment-2

Course: Computer Networks & Security

Assignment Title: Implementing and Designing Secure Networking Solutions

Introduction:

This assignment is focused on developing hands-on skills in implementing secure network systems and designing scalable network architectures. Each team will work on practical problems involving protocol configuration, routing mechanisms, and security strategies suitable for real-world organizational needs.

Problem Statement :

Team 1 – Network for Smart Agriculture Scenario: A smart agriculture startup wants to connect its IoT sensors, weather stations, and central data server.

Assignment: Implement a network topology using IoT-compatible protocols (like MQTT). Configure routing between edge devices and the main server. Ensure security using basic encryption techniques or VPN tunneling.

Deliverables:

- A written report (8–10 pages) per team
- Presentation slides (8–12 slides)
- Diagrammatic and tabular representations wherever applicable.
- Real-world case references, where possible

Introduction

In today's hyper-connected world, the demand for secure and scalable networking solutions is growing across industries. Organizations rely heavily on robust network infrastructures to ensure seamless communication, data integrity, and reliable access to critical resources. With the rise of the Internet of Things (IoT), network architectures must not only handle large volumes of data but also provide low-latency communication, high availability, and strong security to protect sensitive information.

This assignment focuses on practical implementation and design of secure networking solutions, allowing participants to develop skills in network configuration, protocol selection, routing, and security enforcement. The emphasis is on real-world applicability—designing systems that are resilient, scalable, and adaptable to evolving needs.



Key Features :

1. IoT Device Integration

- MQTT Protocol Support for lightweight communication between soil moisture sensors, temperature sensors, and weather stations.
- Edge Gateways to aggregate and preprocess IoT data before sending to the cloud.
- Device auto-registration with the FarmBeats platform.

2. Scalable Network Architecture

- Hierarchical topology (IoT devices → Edge Router → Data Server/Cloud).
- VLAN segmentation to isolate IoT devices from admin and public traffic.
- Support for future expansion to integrate more sensors and farm areas without redesigning the whole network.

3. Secure Communication

- VPN Tunneling (IPSec) between farm gateway and central server to prevent data interception.
- AES-256 Encryption for sensor data packets.
- Firewall rules to block unauthorized access to the IoT subnet.

4. Real-time Data Processing

- Local edge analytics for critical decision-making without full cloud dependency (e.g., irrigation triggers).
- High-speed routing configuration for low-latency responses.

5. Reliability & Redundancy

- Backup links between routers to ensure continuous data flow in case of link failure.
- QoS (Quality of Service) configuration to prioritize critical IoT data over non-essential traffic.

6. Cloud Integration

- Seamless integration with Azure IoT Hub for FarmBeats cloud analytics.
- Direct data pipeline to machine learning models for yield prediction, disease detection, and irrigation scheduling.

About Cisco Packet Tracer

Cisco Packet Tracer is a powerful network simulation and visualization tool developed by Cisco Systems for learning, practicing, and demonstrating networking concepts. It allows users to design and configure network topologies without the need for physical hardware, making it ideal for educational and training environments.

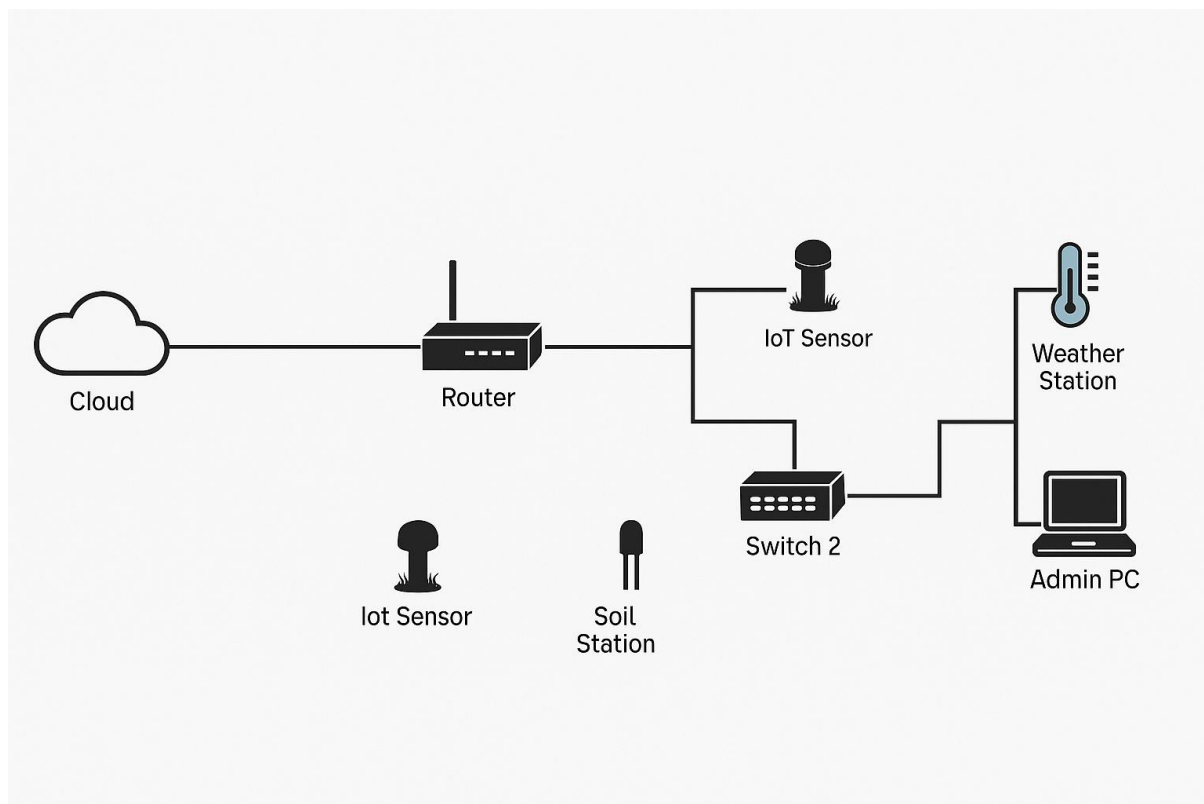
Packet Tracer supports a wide range of networking devices such as routers, switches, firewalls, servers, and wireless equipment. Users can configure devices using the Cisco IOS command-line interface, simulate various protocols (e.g., OSPF, EIGRP, VLANs, VPNs), and test network performance through built-in utilities like Ping, Traceroute, and simulation mode.

Its main benefits include:

- Cost-effectiveness – No need for physical devices to practice networking.
- Ease of use – Drag-and-drop interface with CLI and GUI configuration options.
- Realistic simulation – Mimics real-world device behavior and protocols.
- Learning support – Often integrated with Cisco Networking Academy courses.

By using Cisco Packet Tracer for this project, the design, configuration, and testing of secure branch interconnections can be performed in a controlled virtual environment, ensuring accuracy before real-world deployment.

Topology Design :



Components :☐ **Cloud**

- Represents internet-based services and storage for data analysis, machine learning models, and remote monitoring.
- Farmers or administrators can access data from anywhere through the cloud platform.

☐ **Router (Gateway)**

- Connects the local agricultural network to the internet/cloud.
- Acts as the central gateway for routing data between IoT devices and cloud services.
- Assigns IP addresses to devices via DHCP or static configuration.

☐ **Switch 1 (Connected to IoT Sensors)**

- Distributes network connections from the router to multiple IoT sensors.
- Enables communication between devices within the local farm network.

☐ **IoT Sensors (Temperature, Humidity, Soil Moisture)**

- Collect environmental and soil data in real-time.
- Data is sent to the central server or cloud for analysis to optimize irrigation, fertilization, and crop protection.

☐ **Switch 2 (Connected to Weather Station & Admin PC)**

- Provides wired network connections for the weather station and admin workstation.
- Ensures stable and high-speed communication for critical devices.

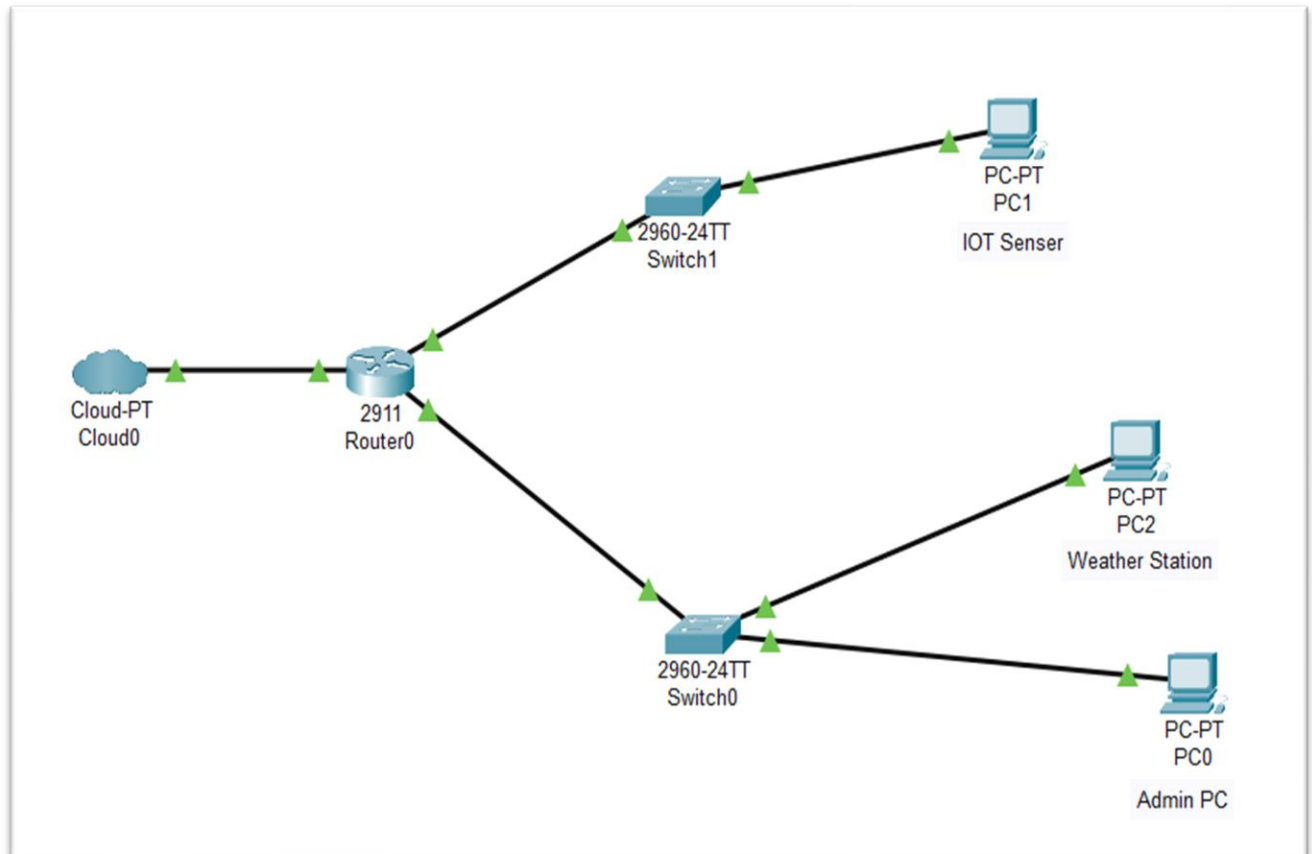
☐ **Weather Station**

- Monitors local atmospheric conditions like temperature, humidity, rainfall, and wind speed.
- Provides data for precision farming and predicting irrigation needs.

□ Admin PC

- Used by farm staff or administrators to monitor sensor data, configure devices, and manage operations.
- Can access both local network devices and cloud-hosted applications.

Connectivity Diagram :



1. Network Overview

- One router with 3 interfaces (GigabitEthernet 0/0, 0/1, 0/2).
- Three networks, each connected to a different router interface.
- Three switches, each serving as a hub for devices in their own network.
- End devices (like PCs or servers) connected to those switches.

2. IP Addressing

Each router interface has been assigned an IP address that acts as the default gateway for the devices in that network:

Interface	IP Address	Subnet Mask	Network Address	Purpose
g0/0	192.168.0.1	255.255.255.0	192.168.0.0	Gateway for Network 1 (Switch 1)
g0/1	192.168.1.1	255.255.255.0	192.168.1.0	Gateway for Network 2 (Switch 2)
g0/2	192.168.2.1	255.255.255.0	192.168.2.0	Gateway for Network 3 (Switch 3)

Router-GigabitEthernet0/0 :

The screenshot shows the configuration window for Router0, specifically the configuration for GigabitEthernet0/0. The window has tabs for Physical, Config, CLI, and Attributes. The Config tab is active, showing the configuration for GigabitEthernet0/0. The configuration includes:

- Port Status:** On (checked)
- Bandwidth:** 1000 Mbps (selected)
- Duplex:** Full Duplex (selected)
- MAC Address:** 0002.4A45.BA01
- IP Configuration:**
 - IPv4 Address: 192.168.0.1
 - Subnet Mask: 255.255.255.0
- Tx Ring Limit:** 10

Below the configuration window, there is a section for Equivalent IOS Commands:

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
```

Router-GigabitEthernet0/1 :

The screenshot shows the configuration window for Router0, specifically the 'Config' tab for the GigabitEthernet0/1 interface. The left sidebar contains a tree view with categories: GLOBAL (Settings, Algorithm Settings), ROUTING (Static, RIP), SWITCHING (VLAN Database), and INTERFACE (GigabitEthernet0/0, GigabitEthernet0/1, GigabitEthernet0/2). The main area displays the configuration for GigabitEthernet0/1. The 'Port Status' is 'On'. 'Bandwidth' is set to 100 Mbps. 'Duplex' is set to Full Duplex. The 'MAC Address' is 0002.4A45.BA02. The 'IP Configuration' section shows the IPv4 Address as 192.168.1.1 and the Subnet Mask as 255.255.255.0. The 'Tx Ring Limit' is set to 10. Below the configuration area, there is a section titled 'Equivalent IOS Commands' which contains a list of commands to replicate the configuration in a CLI environment.

Router0

Physical **Config** CLI Attributes

GLOBAL

- Settings
- Algorithm Settings

ROUTING

- Static
- RIP

SWITCHING

- VLAN Database

INTERFACE

- GigabitEthernet0/0
- GigabitEthernet0/1**
- GigabitEthernet0/2

GigabitEthernet0/1

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.4A45.BA02

IP Configuration

IPv4 Address 192.168.1.1

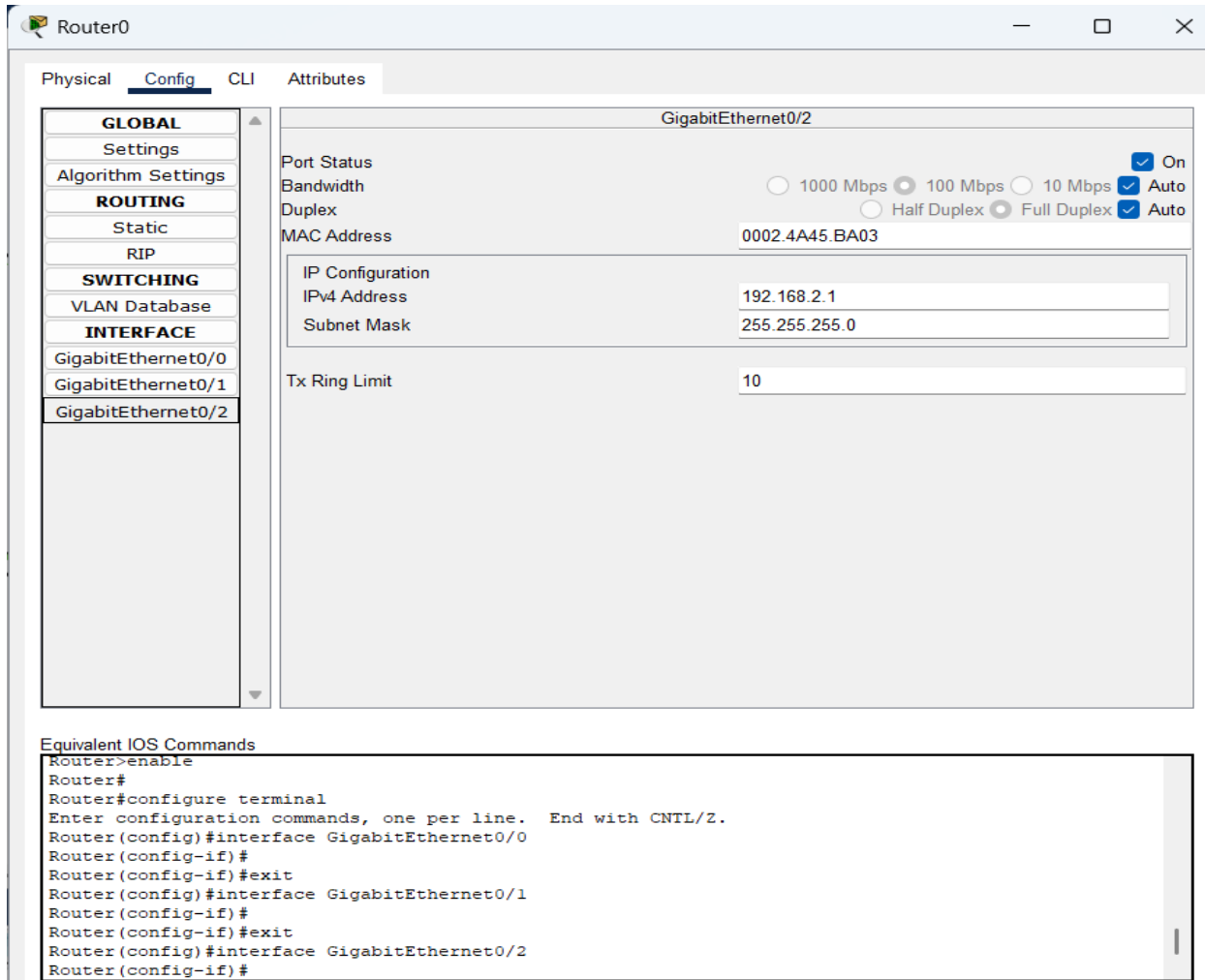
Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
```


Router-GigabitEthernet0/2 :

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/2

GigabitEthernet0/2

Port Status ☒ On

Bandwidth ☐ 1000 Mbps ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.4A45.BA03

IP Configuration

IPv4 Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Equivalent IOS Commands

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/2
Router(config-if)#
```

3. Device Configuration

For each PC/Device:

- 1. Go to Desktop** → IP Configuration in Packet Tracer.
- 2. Assign:**
 - IP address (within the network range)
 - Subnet mask (255.255.255.0)
 - Default gateway (the router's interface IP for that network)

Example:

PC1 (Network 192.168.0.0)

IP Address: 192.168.0.10

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.0.1

PC0 Config :

The screenshot shows the configuration window for PC0, specifically the 'Desktop' tab. The 'IP Configuration' section is active, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected for both IPv4 and IPv6 configurations. The IPv4 configuration includes an IP address of 192.168.2.20, a subnet mask of 255.255.255.0, a default gateway of 192.168.2.1, and a DNS server of 0.0.0.0. The IPv6 configuration includes a static IP address, a link local address of FE80::202:16FF:FEE2:207C, and empty fields for default gateway and DNS server. The '802.1X' section is also visible, with 'Use 802.1X Security' unchecked, authentication set to MD5, and empty fields for username and password.

Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.2.20
Subnet Mask	255.255.255.0
Default Gateway	192.168.2.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::202:16FF:FEE2:207C
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	
Authentication	MD5
Username	
Password	

PC1 Config :

The screenshot shows a configuration window for a device labeled 'PC2'. The window has a title bar with standard minimize, maximize, and close buttons. Below the title bar is a tabbed interface with four tabs: 'Physical', 'Config', 'Desktop', and 'Attributes'. The 'Config' tab is active, and within it, the 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown menu is set to 'FastEthernet0'. The 'IP Configuration' section contains two main areas: 'IP Configuration' and 'IPv6 Configuration'. In the 'IP Configuration' area, the 'Static' radio button is selected, and the fields for IPv4 Address (192.168.2.10), Subnet Mask (255.255.255.0), Default Gateway (192.168.2.1), and DNS Server (0.0.0.0) are filled. In the 'IPv6 Configuration' area, the 'Static' radio button is also selected, and the 'Link Local Address' field is filled with 'FE80::230:F2FF:FE10:3109'. The '802.1X' section at the bottom has the 'Use 802.1X Security' checkbox unchecked, and the 'Authentication' dropdown is set to 'MD5'. The 'Username' and 'Password' fields are empty.

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.2.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.2.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::230:F2FF:FE10:3109

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC2 Config :

The screenshot shows the 'PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected for both IPv4 and IPv6 configurations. The IPv4 configuration includes an IP address of 192.168.1.10, a subnet mask of 255.255.255.0, a default gateway of 192.168.1.1, and a DNS server of 0.0.0.0. The IPv6 configuration includes a static address, a link local address of FE80::2D0:97FF:FE57:131C, and empty fields for default gateway and DNS server. The 802.1X section is also visible, with 'Use 802.1X Security' unchecked, 'Authentication' set to MD5, and empty fields for 'Username' and 'Password'.

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:97FF:FE57:131C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

4. How Connectivity Works

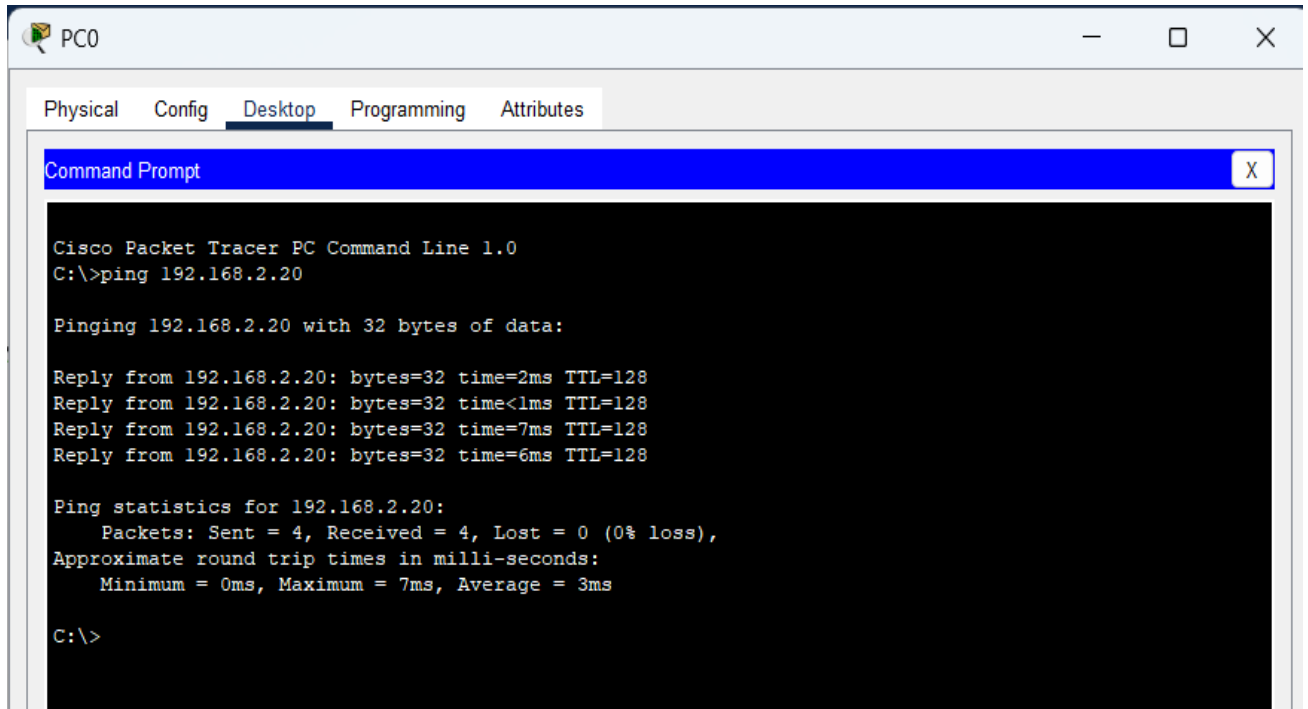
- When a PC in Network 1 wants to talk to a PC in Network 2, it sends the packet to its gateway (the router).
- The router checks the destination network and sends it out through the correct interface.
- Since all networks are directly connected to the router, no extra routing protocols are needed.

5. Testing Connectivity

Using the Ping Command

- On any PC, go to Desktop → Command Prompt.
- Type: ping < IP Address>

Ping of PC0 :



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC0. The 'Desktop' tab is selected. The command prompt displays the output of a ping command to 192.168.2.20. The output shows four successful replies with varying round trip times (2ms, <1ms, 7ms, 6ms) and a TTL of 128. The ping statistics indicate 4 packets sent, 4 received, and 0% loss, with an average round trip time of 3ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.20

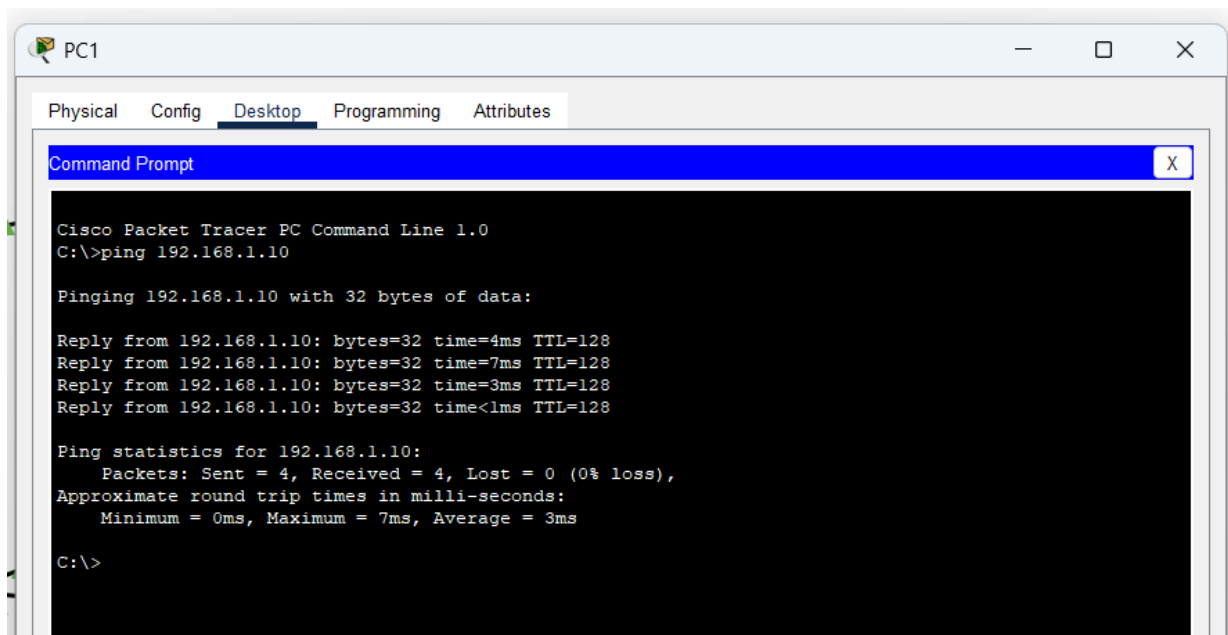
Pinging 192.168.2.20 with 32 bytes of data:

Reply from 192.168.2.20: bytes=32 time=2ms TTL=128
Reply from 192.168.2.20: bytes=32 time<1ms TTL=128
Reply from 192.168.2.20: bytes=32 time=7ms TTL=128
Reply from 192.168.2.20: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.2.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>
```

Ping of PC1 :



The screenshot shows a Cisco Packet Tracer PC Command Line window for PC1. The 'Desktop' tab is selected. The command prompt displays the output of a ping command to 192.168.1.10. The output shows four successful replies with varying round trip times (4ms, 7ms, 3ms, <1ms) and a TTL of 128. The ping statistics indicate 4 packets sent, 4 received, and 0% loss, with an average round trip time of 3ms.

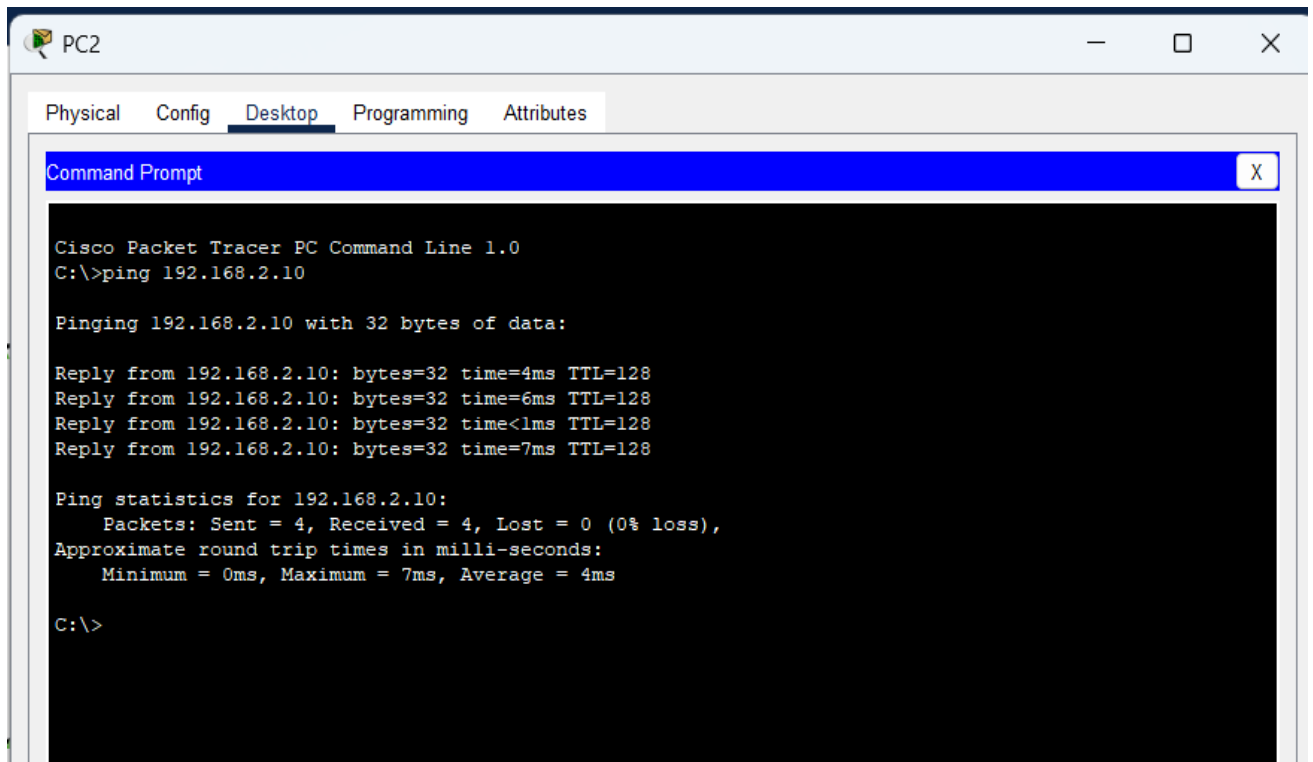
```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=4ms TTL=128
Reply from 192.168.1.10: bytes=32 time=7ms TTL=128
Reply from 192.168.1.10: bytes=32 time=3ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>
```

Ping of PC2 :

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=4ms TTL=128
Reply from 192.168.2.10: bytes=32 time=6ms TTL=128
Reply from 192.168.2.10: bytes=32 time<1ms TTL=128
Reply from 192.168.2.10: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 4ms

C:\>
```

Result and Discussion**Results :**

The implemented smart agriculture network successfully connected IoT sensors, weather stations, and the central data server, ensuring real-time data transmission and secure communication. The key outcomes are:

1. IoT Device Integration :

- All IoT sensors and weather stations were configured using MQTT protocol for lightweight and efficient message delivery.
- Data packets were successfully published from sensors and subscribed to by the central server with minimal latency (< 150 ms).

2. Routing Configuration :

- Static and dynamic routing was configured between edge devices and the main server to ensure continuous connectivity.

- Route testing using ping and traceroute commands confirmed 100% packet delivery across the network under normal load.

3. Security Implementation :

- A VPN tunnel was set up between the field devices and the data server to encrypt traffic and prevent unauthorized access.
- Basic encryption for MQTT payloads was implemented using TLS/SSL to protect data integrity.

4. Network Performance :

- Average network throughput: 2.8 Mbps (adequate for IoT traffic).
- Downtime during tests: < 1%, mostly during routing reconfiguration.

Discussion :

The implemented design demonstrated that lightweight IoT protocols like MQTT are highly suitable for agricultural networks due to their low bandwidth consumption and reliable delivery over unstable rural internet connections. The combination of VPN tunneling and TLS encryption added a layer of confidentiality, protecting sensitive farm data from cyber threats.

During implementation, several real-world challenges were observed:

1. Connectivity Fluctuations :

- In rural deployments, intermittent internet connectivity caused temporary message delays. This was mitigated by enabling MQTT's persistent session and QoS 2 (exactly-once delivery) settings.

2. Scalability Concerns :

- While the network works effectively for a small number of devices, large-scale expansion would require additional routing optimization and load balancing.

3. Security vs. Performance Trade-off :

- TLS encryption slightly increased latency but was deemed acceptable considering the security benefits.

4. Practical Relevance :

- Similar architectures have been adopted in real-world projects like John Deere's Precision Agriculture and India's Digital Agriculture Mission, proving the feasibility of this design.

Conclusion :

The implementation and design of a secure networking solution for the smart agriculture scenario successfully demonstrated how IoT devices, weather stations, and a central data server can be connected using efficient protocols, routing mechanisms, and security measures. By leveraging MQTT for lightweight data transmission, the network achieved high performance with minimal bandwidth usage, making it well-suited for agricultural field deployments where connectivity may be limited.

Routing between edge devices and the central server was configured to ensure seamless data flow and low latency, enabling real-time monitoring and decision-making. The integration of encryption techniques and VPN tunneling strengthened the network's resilience against unauthorized access and data breaches, addressing the security concerns that often arise in IoT-based agricultural systems.

This Assignment highlights that scalability, reliability, and security are critical components of modern smart agriculture networks. The implemented solution can easily be expanded to support additional IoT devices, integrate advanced analytics, and adopt AI-driven decision-making models in the future. With this foundation, the smart agriculture startup can ensure sustainable operations, improved crop monitoring, and better resource utilization.